

Developmental exposure to glyphosate-based herbicide and depressive-like behavior in adult offspring: Implication of glutamate excitotoxicity and oxidative stress

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ABSTRACT

We have previously demonstrated that maternal exposure to glyphosate-based herbicide (GBH) leads to glutamate excitotoxicity in 15-day-old rat hippocampus. The present study was conducted in order to investigate the effects of subchronic exposure to GBH on some neurochemical and behavioral parameters in immature and adult offspring. Rats were exposed to 1% GBH in drinking water (corresponding to 0.36% of glyphosate) from gestational day 5 until postnatal day (PND)-15 or PND60. Results showed that GBH exposure during both prenatal and postnatal periods causes oxidative stress, affects cholinergic and glutamatergic neurotransmission in offspring hippocampus from immature and adult rats. The subchronic exposure to the pesticide decreased L-[¹⁴C]-glutamate uptake and increased ⁴⁵Ca²⁺ influx in 60-day-old rat hippocampus, suggesting a persistent glutamate excitotoxicity from developmental period (PND15) to adulthood (PND60). Moreover, GBH exposure alters the serum levels of the astrocytic protein S100B. The effects of GBH exposure were associated with oxidative stress and depressive-like behavior in offspring on PND60, as demonstrated by the prolonged immobility time and decreased time of climbing observed in forced swimming test. The mechanisms underlying the GBH-induced neurotoxicity involve the NMDA receptor activation, impairment of cholinergic transmission, astrocyte dysfunction, ERK1/2 overactivation, decreased p65 NF-κB phosphorylation, which are associated with oxidative stress and glutamate excitotoxicity. These neurochemical events may contribute, at least in part, to the depressive-like behavior observed in adult offspring.

1. Introduction

Glyphosate (*N*-phosphonomethyl-glycine) is an organophosphorus compound, widely used as a broad-spectrum, post-emergent, non-selective herbicide, as well as a crop desiccant. It has been used in both agricultural and non-agricultural activities (e.g. gardening). Commercial formulations containing glyphosate account for approximately 60% of the world market of non-selective herbicides (IBAMA, 2014; IARC, 2015). In Brazil, the planting of glyphosate-resistant soybeans (GRS) has greatly enhanced the consumption of glyphosate formulations in crops.

Glyphosate-based herbicides (GBH) are frequently formulated as isopropylamine salts and surfactants may be added to formulations (FAO, 2000/2001; IPCS, 1994). The use of polyethoxylated tallowamine (POEA) as a surfactant in GBH formulations facilitate the herbicide uptake by plants (Székács and Darvas, 2012). It has been suggested that formulations containing surfactants may be even more toxic than the glyphosate alone to mammal cells (de Liz Oliveira Cavalli et al., 2013; Mesnage et al., 2013; Sribanditmongkol et al., 2012).

In 2000, Williams and coworkers reviewed the data concerning the safety of GBH to human and animal health. They suggested that the use of these herbicides did not cause adverse effects to mammals (Williams

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et al., 2000). However, in a recently statement of concerns over the use of GBH and risks associated with exposures, Myers et al. (2016) reported a considerable number of animal and epidemiology studies published in the last decade demonstrating GBH toxicity to animals and humans. Based on these data, the authors concluded that human exposures to glyphosate are rising and the current safety standards for GBH are outdated and may fail to protect public health or the environment.

The presence of glyphosate and its main metabolite AMPA (aminomethylphosphonic acid) are generally detected together and more frequently in soil and sediments, ditches and sewers, rivers and streams (Contardo-Jara et al., 2009; Aparicio et al., 2013; Battaglin et al., 2017). Thus, non-target organisms may be exposed to glyphosate through drinking water (Larsen et al., 2012; Mbanaso et al., 2014). Indeed, evidences of human exposure to this herbicide has been shown through the detection of glyphosate in urine samples of people living in farm and non farm household (Acquavella et al., 2004; Conrad et al., 2017).

A recent study also demonstrated the presence of glyphosate in maternal and umbilical cord serum in pregnant women who gave birth in three provinces of Thailand (Kongtip et al., 2017). The authors showed that pregnant women who work in agriculture or live with families that work in agriculture had higher exposures to the herbicide glyphosate when compared to pregnant women who were not agriculturists/farmers by occupation. In this context, studies involving the transport kinetics of glyphosate across the BeWo cell monolayer (used as model of human trophoblast) and across the placental barrier in the *ex vivo* perfusion system showed that the transport of glyphosate through placenta is possible, even though is considered low. After 2.5 h of perfusion, the percentage of compound transferred to the fetal compartment was about 15% for glyphosate (Mose et al., 2008) and after 24 h of BeWo cell experiments, the percentage of glyphosate transported was near 24% (Poulsen et al., 2009).

Several studies have demonstrated that pre- and postnatal exposure to different types of pesticides may be associated with neurological and neuropsychiatric effects. Even at low concentrations, exposure to neurotoxins during central nervous system (CNS) development can promotes injuries to brain areas like prefrontal cortex and hippocampus, causing losses in cognitive functions (Ainge et al., 2007; Chen et al., 2012; Mogensen et al., 2007). This exposure to neurotoxicants can be very subtle and became more evident only in older ages (Bondy and Campbell, 2005; Rice and Barone, 2000).

Utilizing the zebrafish vertebrate model system to study early effects of glyphosate and GBH exposure, Roy et al. (2016) found morphological abnormalities in the developing brain as well as a decrease in genes expressed in different regions of the brain. Another study using zebrafish model demonstrated an impaired neuronal development caused by glyphosate exposure during the embryonic development period. The exposure led to an inhibition of carbonic anhydrase activity with production of ROS especially in branchial regions, triggered cellular apoptosis resulting in several types of malformations (Sulukan et al., 2017). It was recently demonstrated that initial axonal differentiation and growth of cultured neurons is affected by glyphosate. Biochemical approaches revealed that glyphosate led to a decrease in Wnt5a level, a key factor for the initial neurite development and maturation, as well as inducing a down-regulation of CaMKII activity. Additionally, these changes might be reflected in a subsequent neuronal dysfunction (Coullery et al., 2016).

In our previously work regarding maternal exposure to GBH we demonstrated that exposure during pre and postnatal periods leads to calcium overload and glutamate excitotoxicity in immature offspring hippocampus (Cattani et al., 2014). The mechanism underlying such effects involves the calcium influx by activating *N*-methyl-D-aspartate (NMDA) glutamate receptors and voltage-dependent calcium channels, which leads to oxidative stress and neural cell death in immature rat hippocampus (PND15). The mechanism involved in NMDA receptor

overactivation by glyphosate is not elucidated. The glutamate excitotoxicity with subsequent intracellular calcium influx is one of the main factors for reactive oxygen species (ROS) generation in brain (Emerit et al., 2004; Gilgun-Sherki et al., 2001; Kaur and Ling, 2008; Zamoner et al., 2008), leading to oxidative damage.

The present study was conducted in order to evaluate if these effects of maternal exposure to GBH during gestational and suckling periods, previously observed in immature offspring hippocampus (PND15) (Cattani et al., 2014), persist until adulthood (postnatal day 60, PND60). In this context, several mechanistic/neurochemical endpoints were evaluated in this follow-up study, as well as the behavioral alterations induced by the subchronic herbicide exposure in adult animals, trying to provide benchmark data for a risk assessment for GBH to neural cells. Moreover, given the similarity of glyphosate structure to glycine and glutamate we investigated whether glyphosate might affect the glutamate neurotransmission system by using molecular docking of glyphosate to glycine and glutamate binding sites on the NMDA glutamate receptors.

2. Material and methods

2.1. Radiochemical and compounds

L-[¹⁴C]-glutamate (specific activity 9.62 GBq/mmol) was purchased from Amersham (Oakville, Ontario, Canada). [⁴⁵Ca²⁺]-CaCl₂ (specific activity of 321 kBq/mg of Ca²⁺) and Optiphase Hisafe III biodegradable liquid scintillation were purchased from Perkin Elmer (Waltham, MA). Anti-p44/42 MAP Kinase (anti-ERK1/2), anti-phospho-p44/42 MAP kinase (anti-phospho ERK1/2), antibodies were from Cell Signaling Technology, Inc. (USA). The herbicide Roundup Original® (Homologation number 00898793) containing glyphosate 360 g/L is a commercial formulation registered in the Brazilian Ministry of Agriculture, Livestock and Supply (Ministério da Agricultura, Pecuária e Abastecimento – MAPA). The Immobilon™ Western chemiluminescent HRP substrate was obtained from Millipore. The NF-κB p65 (Total) InstantOne ELISA and InstantOnePhospho-NF-κB p65 (Ser536) ELISA kits were purchased from eBioscience® (San Diego, CA, USA). All other chemicals were of analytical grade.

2.2. Animals

Wistar rats were bred in animal facility and maintained in an air-conditioned room (21 ± 1 °C) with controlled lighting (12 h/12 h light/dark cycle). On the day of birth, the litter size was culled to eight pups (4 males and 4 females, whenever possible). Litters smaller than eight pups were not included. Pelleted food (Nuvital, Nuvilab CR1, Curitiba, PR, Brazil) and tap water were available *ad libitum*. All animals were observed for clinical signs of toxicity related to chemical treatment. All experimental procedures were conducted in accordance to the rules and laws of ethical recommendations of the local Ethical Committee for Animal Use at Federal University of Santa Catarina (CEUA/UFSC, Protocol #PP00820), which is in accordance with Brazilian Council for Control of Animal Experimentation (CONCEA).

2.3. Animal treatment

The experimental protocol of maternal exposure to GBH used herein was previously described by us (Cattani et al., 2014). Wistar rats were mated and the day of appearance of the vaginal plug was considered gestational day 0 (GD0). Pregnant females were housed individually and divided into 2 groups (4 dams in each group): control group provided with tap water and GBH group provided with 1% GBH in drinking water (corresponding to 0.36% of glyphosate). Maternal exposure to tap water or GBH-containing drinking water was started at gestational day 5 (GD5) and continually up to postnatal day 15 (PND15) or up to postnatal day 60 (PND60). Thus, the offspring were exposed *in utero*,

then from PND1 to PND21 via lactation, and at PND21 pups were weaned. After weaning male offspring were housed in groups of four rats per cage receiving 1% GBH (treated group) or tap water (controls) until PND60. The male pups were used to investigate the neurotoxic effects of GBH in hippocampal cells on PND15 and PND60. Each experimental group contained hippocampal slices from rats of different litters (different dams). The neurochemical tests were performed on PND15 and PND60, while the neurobehavioral tests were performed only on PND60. The number of animals used in each test was at least 6 per experimental group. The GBH solutions were prepared daily. Drink consumption and body weight of male offspring were recorded on a daily basis. The maternal dose of GBH was equivalent to 70 mg of glyphosate/Kg/day. This glyphosate dose is lower than the limits of no observed adverse effect level (NOAEL) for maternal toxicity, which is 1000 mg/kg body mass/day dosage, as previously published (Williams et al., 2000). Moreover, the same review demonstrated that the NOAELs in the subchronic rodent toxicity studies ranged from 209 to 1445 mg/kg body weight/day.

The dose used in our experimental protocol aims to assess the potential effects associated with GBH exposure by using animal model providing the scientific community with insight into glyphosate sub-chronic toxicity, covering maternal exposure period. The present study evaluated if the exposure to GBH during development may alter some behavioral aspects during adulthood, as well as if the neurotoxicity previously described in immature rat hippocampus (Cattani et al., 2014) persists until adult age.

2.4. $^{45}\text{Ca}^{2+}$ uptake

Hippocampal slices from PND60 male rats were preincubated in Hank's balanced salt solution (HBSS) containing 16.67 mM NaHCO_3 , 1.29 mM CaCl_2 , 136.9 mM NaCl , 5.36 mM KCl , 0.65 mM MgSO_4 , 0.27 mM Na_2HPO_4 , 6.1 mM glucose, 0.441 mM KH_2PO_4 for 15 min in a Dubnoff metabolic incubator at 37 °C, pH 7.4. After that, the medium was changed by fresh HBSS with 0.1 $\mu\text{Ci}/\text{mL}$ $^{45}\text{Ca}^{2+}$ and incubated during 30 min. Extracellular $^{45}\text{Ca}^{2+}$ was thoroughly washed off in a washing solution containing 127.5 mM NaCl , 4.6 mM KCl , 1.2 mM MgSO_4 , 10 mM HEPES, 11 mM glucose, 10 mM LaCl_3 , pH 7.3 (30 min in washing solution). The presence of La^{3+} during the washing stage was used to prevent release of the intracellular $^{45}\text{Ca}^{2+}$ (Zamoner et al., 2007). After washing, tissue slices were digested and homogenized with 0.5 M NaOH solution, 100 μL aliquots were placed in scintillation fluid and counted in a LKB Beck beta liquid scintillation spectrometer (model LS 6500; Multi-Purpose Scintillation Counter-Beckman Coulter, Boston, USA), and 5 μL aliquots were used for protein quantification.

2.5. ^3H -glutamate uptake

The ^3H -glutamate uptake was performed in hippocampal slices from offspring at PND60. For this, the incubation medium was replaced by Hank's balanced salt solution (HBSS) containing 137 mM NaCl ; 0.27 mM Na_2HPO_4 ; 4.17 mM NaHCO_3 ; 5.36 mM KCl ; 1.1 mM KH_2PO_4 ; 1.29 mM CaCl_2 ; 0.65 mM MgSO_4 ; 0.49 mM MgCl_2 , 2 mM glucose and 5 mM HEPES pH 7.4. Then, glutamate uptake assay was started by the addition of 0.33 $\mu\text{Ci}/\text{mL}$ ^3H -glutamate. Incubation was stopped after 10 min by removal of the medium and rinsing the slices twice with ice-cold HBSS. Slices were then lysed in a solution containing 0.5 M NaOH . Sodium-independent uptake was determined using N-methyl-D-glucamine instead of sodium chloride. Sodium-dependent glutamate uptake was obtained by subtracting the non-specific uptake from the specific uptake (Cesconetto et al., 2016). Radioactivity was measured with a scintillation counter.

2.6. Determination of cholinesterase activity

The hippocampus from PND15 or PND60 offspring was removed

and homogenized in 50 mM Tris-HCl buffer pH 7.4. Then cholinesterase activity was measured using propionylthiocholine as a substrate. The rate of hydrolysis of propionylthiocholine was measured at 412 nm through the release of the thiol compound which, when reacted with 5,5'-dithiobis-(2-nitrobenzoic acid) (DTNB), produces the color-forming compound TNB. This methodology was also used to determine the cholinesterase activity in serum from the PND15 or PND60 offspring.

2.7. Enzymatic activity of GGT

Gamma-glutamyl transferase (GGT) activity was measured in hippocampus homogenate from rats on PND15 or PND60 with the use of the modified technique described previously by Orłowski and Meister (1963), using L- γ -glutamyl p-nitroanilide as substrate and glycylglycine as the acceptor molecule. Aliquots of the tissue homogenate were incubated with the enzymatic substrate. The reaction was allowed to proceed for 60 min at 37 °C and the enzymatic reaction was stopped by addition of acetic acid. The absorbance of the samples was determined in a plate reader (Tecan Infinite[®] 200 PRO) at 530 nm. The results were expressed as $\text{U L}^{-1} \cdot \mu\text{g protein}^{-1}$.

2.8. S100B and NF- κB measurements

2.8.1. S100B

The S100 B immunocontent was determined by ELISA in samples serum from rats on PND15 or PND60. ELISA for S100 B was carried out as previously described (Leite et al., 2008).

2.8.2. Measurement of total and phosphorylated levels of NF- κB

To analyze the effect of GBH exposure on phosphorylation of the p65 protein (subunit of NF- κB), the total and phosphorylated levels of p65 (p65 and p-p65, respectively) were measured. Rat hippocampus from PND15 was collected, homogenized in lysis buffer and protein concentration was determined (Lowry et al., 1951). Total protein content was adjusted and we used 60 $\mu\text{g}/\text{sample}$. The analysis was performed using a commercially available kit with specific monoclonal antibodies against total p65 protein and specifically against phosphorylated p65 protein. The experiment was carried out using the NF- κB p65 (Total) InstantOne ELISA and InstantOnePhospho-NF- κB p65 (Ser536) ELISA Kits (purchased from eBioscience[®], San Diego, CA, USA) according to the manufacturer's protocols. The results are expressed as the fold change relative to the (S) group, which represents the basal total and phosphorylated p65 levels.

2.9. Antioxidants enzyme assays

To determine CAT, SOD, GR, GPx, GST and G6PD activities, hippocampus from control or GBH-treated groups from PND15 or PND60 offspring were homogenized in a cold buffer containing 20 mM sodium phosphate, pH 7.4 0.1% Triton, and 150 mM NaCl (1:20 w/v). The determinations were performed using the supernatant after centrifugation of the homogenate ($5000 \times g$ for 5 min).

2.9.1. Catalase activity

The CAT activity was determined in tissue homogenates according to the method previously described (Aebi, 1984), based on measuring the decreased absorbance in a freshly prepared 10 mM hydrogen peroxide solution at 240 nm for 30s. The enzyme activity was expressed as $\text{mmol g}^{-1} \text{min}^{-1}$.

2.9.2. Superoxide dismutase activity

The SOD activity measurement is based on epinephrine oxidation (pH 2.0 to pH 10.2), which produces superoxide anion and adrenochrome, which is measured at 480 nm (Misra and Fridovich, 1972; Boveris et al., 1983). A unit of SOD is defined as the amount of enzyme that inhibits the speed of adrenalin oxidation by 50% and the result

expressed in U mL^{-1} .

2.9.3. Glutathione peroxidase activity

The GPx activity was analyzed by the method based on *tert*-butyl hydroperoxide reduction via oxidation of GSH to GSSG, catalyzed by GPx, and subsequent regeneration of GSH by the enzyme GR with oxidation of NADPH at 340 nm. Therefore, the rate of oxidation of NADPH is proportional to the activity of the GPx in the sample (Flohé and Günzler, 1984). The enzyme activity was expressed as $\mu\text{mol g}^{-1} \text{min}^{-1}$.

2.9.4. Glutathione reductase activity

The GR activity was determined by measuring the rate of NADPH oxidation at 340 nm due to the formation of GSH, from GSSG, by the action of GR present in the sample (Carlberg and Mannervik, 1985). The unit of enzyme activity was expressed as $\mu\text{mol g}^{-1} \text{min}^{-1}$.

2.9.5. Glutathione S-transferase activity

The GST activity was measured using the methodology described by Habig et al. (1974). In this protocol, 1-chloro-2,4-dinitrobenzene (CDNB; as substrate for GST) was used, the enzyme promoting the CDNB–GSH conjugation. This reaction was spectrophotometrically monitored for 60 s at 340 nm. The enzyme activity was expressed as $\mu\text{mol g}^{-1} \text{min}^{-1}$.

2.9.6. Glucose-6-phosphate dehydrogenase assay

For measuring the G6PD activity, aliquots of tissue homogenate were incubated in the presence of NADP^+ leading to the oxidation of glucose-6-phosphate to 6-phosphogluconate. The NADPH produced was measured in a kinetic mode during 10 min. The results were calculated by assessing the increase of the optical density per minute (slope) of the sample against the “slope” of standard G6PD enzyme activity. The G6PD assay kit was kindly provided by Intercientífica (São José dos Campos, SP, Brazil).

2.10. Reduced glutathione assay

The reduced glutathione (GSH) was determined according to Beutler et al. (1963) in hippocampus from PND15 or PND60 offspring using the reagent DTNB (5,5'-dithiobis 2-nitrobenzoic acid). After being centrifuged at 5000g for 5 min, the supernatants from the acid extracts (TCA 12%, 1:10, w/v) were added to 2.5 mM DTNB in 0.2 M sodium phosphate buffer pH 8.0, and the formation of the thiolate anion was immediately measured at 412 nm. Determinations were expressed in $\mu\text{mol g}^{-1}$.

2.11. Behavioral tests

The animals were kept until PND60 under GBH exposure. All animals were tested at PND60 on a forced swimming, open field, rotarod and sucrose consumption tests. Animals were habituated to the test room for 1 h prior to the initiation of the behavior tests, which were carried out during the light phase of the cycle (10:00–17:00 h).

2.11.1. Forced swimming test

The forced swimming test were carried out on rats according to the method of Porsolt et al. (1978). 60-days old rats were individually forced to swim in open plastic cylinders (height 60 cm, diameter 30 cm) containing 40 cm of water, maintained at $25^\circ\text{C} \pm 1^\circ\text{C}$. Two swimming sessions were conducted (an initial 15-min training followed 24 h later by a 5-min test). The total duration of immobility and of climbing during the 5-min test session were scored. A digital video camera mounted over the tank recorded behaviors during the test. The following measures were scored on a video monitor from taped: (i) immobility time: the animal was completely motionless, no movements of limb, tail, or head; (ii) climbing time: front paw movements against the

tank wall bringing part of the body out of the water. Immediately after the test, each animal was removed from the water, towel-dried, and returned to its home cage. The water of tank was changed and the cylinder rinsed with clean water after each rat.

2.11.2. Open-field test

To evaluate possible locomotor changes after GBH exposure we used the model proposed by Hall and Ballachey (1932). The apparatus, made of wood covered with impermeable Formica, had a gray floor of 100×100 cm and walls, 40 cm high. Each rat was placed in the center of the apparatus and the total distance traveled (m) and the time in the central region were registered for 15 min using the ANY-maze® video tracking system (Stoelting Co., Wood Dale IL, USA). The apparatus was cleaned with 10% ethanol between animals.

2.11.3. Rotarod test

The rotarod system was used to evaluate motor coordination and balance of the animals. The PND60 rats were placed on the horizontal rotating bar of the rotarod apparatus (Rotamex-V-EE/85) controlled by a computational system (Columbus Instruments Computer-Counter Interface; Ohio, USA). The apparatus consists of a rotating bar, which can rotate with constant speed or with adjustable acceleration (1 rpm/s). The animals were conditioned at a constant speed of 5 rpm for a period of 5 min and the latency to fall (in s) was determined (Shiotsuki et al., 2010).

2.11.4. Sucrose consumption

Sucrose consumption is frequently used as a measure of anhedonia induced by stress in rodents (Willner et al., 1987; Willner, 2005). The PND60 rats were transferred into single housing cages with free access to food. Each rat was provided with two water bottles on the extreme sides of the cage during 24 h training phase to adapt the rats to drinking from two bottles. After training, one bottle was randomly switched for another that contain 0.8% sucrose solution, and 24 h later, the bottles were reversed to avoid perseveration effects. The consumption of water and sucrose solution was estimated simultaneously in groups by weighing the bottles.

2.12. Molecular docking and dynamics simulations

Autodock vina software (Trott and Olson, 2010) was used for molecular docking. The NMDA receptor was prepared for docking using Chimera 10.1 (Pettersen et al., 2004) at default parameters and AMBER force field (Wang et al., 2004). Glyphosate and Glutamic acid ligands were selected from Zinc database (Irwin and Shoichet, 2005). Glyphosate used was modeled with net charge -2 at pH 7. Flexibly residues, 3 Å from ligand bound center (Glutamic acid or glycine pockets), were used for running dock analysis. The lower ΔG pose with lower RMSD to the crystallographic ligand was selected for molecular dynamics simulations (MD). Selected glyphosate binding poses from docking analysis using 4NF8 PDB structure were submitted to 12 ns Molecular Dynamics simulations (Glyphosate at Glycine binding site and Glyphosate at Glutamate binding site). The simulations were performed using GROMACS 5.1.4 (Van Der Spoel et al., 2005). GROMOS96 43a1 force field was used and ligand topology was prepared by PRODRG 2.5 (Schüttelkopf and van Aalten, 2004). The time step for all simulations was 2 fs. First, the complex was solvated in a cubic box using the SPC water model and the system was neutralized by the addition of counter ions. After energy minimization, two 500 ps equilibrations MD were performed under NVT and NPT conditions, respectively. For the equilibrations runs, position restraints were applied to the protein and ligand backbones. Finally, 15 ns production MD were performed under NPT conditions ($T = 300$ K, $P = 1$ bar). Atom coordinates were saved every 10 ps. Binding energies in kJ/mol (electrostatic, polar solvation, Van der Waals and SASA) were calculated by using MM-PBSA tool (Kumari et al., 2014) in each 500 ps from dynamics. The default

parameters were used in all instances. Ligand binding analysis after 12 ns dynamics was analyzed by LigPlot software (Wallace et al., 1995). PyMol (www.pymol.org) was used to create images.

2.13. Western blot analysis

Hippocampus from PND15 was homogenized in 300 μ L of a lysis solution containing 2 mM EDTA, 50 mM Tris-HCl, pH 6.8, 4% (w/v) SDS and protein concentration was determined using serum bovine albumin as the standard (Lowry et al., 1951). Then, samples were dissolved in 25% (v/v) of a solution containing: 40% glycerol, 5% mercaptoethanol, 50 mM Tris-HCl, pH 6.8 and boiled for 3 min. Equal protein concentrations were loaded onto 10% polyacrylamide gels and analyzed by SDS-PAGE and transferred to nitrocellulose membranes for 1 h at 15 V in transfer buffer (48 mM Trizma, 39 mM glycine, 20% methanol and 0.25% SDS). The nitrocellulose membranes were washed for 10 min in Tris-buffered saline (TBS; 0.5 M NaCl, 20 mM Trizma, pH 7.5), followed by 2 h incubation in blocking solution (TBS plus 5% defatted dried milk). After incubation, the blot was washed twice for 5 min with TBS plus 0.05% Tween-20 (T-TBS), and then incubated overnight at 4 °C in blocking solution containing the following antibodies: anti-ERK1/2, anti-phospho ERK1/2 (diluted 1:2000). The blot was then washed twice for 5 min with T-TBS and incubated for 2 h in TBS containing peroxidase conjugated anti-rabbit IgG 1:2000. In addition, we used the following controls in the antibody experiments: primary antibody only, secondary antibody only and negative control (lacking the sample containing the antigen of interest). These controls stated the specificity and sensibility of the antibody. The blot was washed twice again for 5 min with T-TBS and twice for 5 min with TBS. The blot was then developed using the Immobilon™ Western chemiluminescent HRP substrate. Western blots were quantified by scanning the films and determining optical densities with an OptiQuant version 02.00 software (Packard Instrument Company).

2.14. Statistical analysis

All experimental data are represented as mean $\pm$ S.E.M. Black and gray bar graphs represent results obtained from PND15 and PND60 offspring hippocampus, respectively. Statistically significant differences from controls were determined by unpaired Student's *t*-test, unpaired *t*-test with Welch's correction, one-way ANOVA and *post hoc* tests with multiple comparisons (Bonferroni) posttests or two-way ANOVA using GraphPad InStat version 3.01 and GraphPad Prism version 5.0, GraphPad software Inc, (La Jolla, California, USA). Differences were considered to be significant when $p < 0.05$.

3. Results

3.1. Effect of exposure to GBH on body mass gain and offspring well-being

The experimental protocol used in the present study was designed to avoid or minimize both pain and distress in animals. All animals were monitored daily during the treatment and we did not observe any signs of pain and/or distress such as ataxia, decreased activity, eating of bedding or neonates, excessive licking and scratching, loss of appetite, as well as piloerection. Moreover, we did not observe visible external malformations in any of the groups, and the average litter size among dams was not affected by GBH treatment (Table 1). In addition, the GBH exposure did not caused mortality within-litter. On the other hand, two-way ANOVA analysis showed that the offspring body mass gain was decreased by GBH exposure (Fig. 1).

3.2. Involvement of Ca^{2+} and glutamate on the mechanism of GBH-induced neurotoxicity

Previously, we showed that maternal exposure to GBH decreased

Table 1

Average litter size of control and GBH-treated dams.

	Litter size	
	Control	GBH
Male (N)	5.0 $\pm$ 3.21	4.38 $\pm$ 1.85
Female (N)	5.12 $\pm$ 2.23	4.5 $\pm$ 2.45
Total (N)	10.0 $\pm$ 3.54	8.90 $\pm$ 3.14

The litter size was expressed as mean $\pm$ S.D.a.

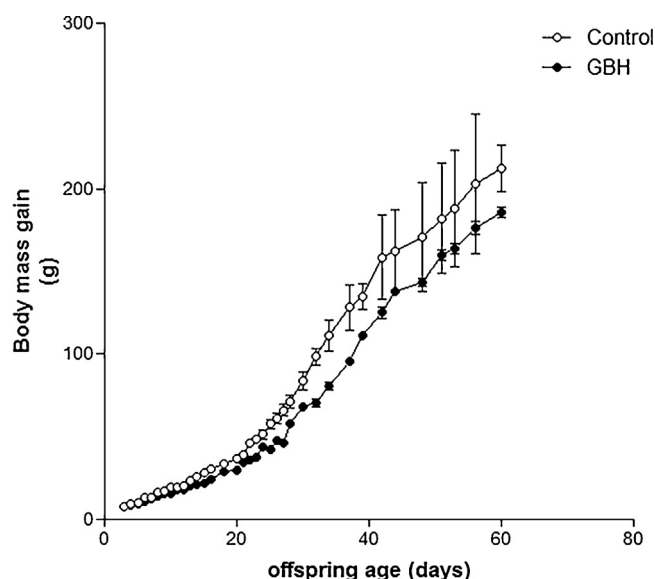


Fig. 1. Effects of developmental exposure to GBH on male offspring body mass gain. Data are reported as means $\pm$ S.D. for 15 animals in each group and analyzed by two-way ANOVA ($P < 0.0001$; $F = 187.1$, for treatment).

glutamate uptake and increased Ca^{2+} influx into hippocampal cells on PND15 (Cattani et al., 2014). The present results showed that subchronic exposure to GBH also provoked decreased Na^{+} -dependent glial [^{14}C]-glutamate uptake (Fig. 2A) in adult rat hippocampus (PND60), suggesting excess of glutamate in synaptic cleft, which may account for glutamate excitotoxicity. The increased extracellular levels of glutamate may activate NMDA receptors allowing a voltage-dependent flow of Ca^{2+} into the cell. Corroborating this, Fig. 2B showed increased $^{45}Ca^{2+}$ uptake induced by GBH subchronic exposure. The excess of glutamate elicited by the decreased uptake of this neurotransmitter supported the misregulated glutamate homeostasis. The present study was also conducted to assess the activities of glutamate metabolism-related enzymes, including aspartate aminotransferase (AST) and alanine aminotransferase (ALT), in the hippocampus of PND60 rats. Results showed that glutamate transamination is not altered by the herbicide exposure (Data not shown).

3.3. Molecular docking and dynamic simulation of glyphosate to NMDA receptor

NMDA receptors play important roles in synaptic plasticity, memory and learning. We previously demonstrated, by using pharmacological tools, that the mechanism of glyphosate-induced calcium influx in rat hippocampus occurred through overstimulation of NMDA receptors (Cattani et al., 2014). These receptors are activated when glutamate and glycine are bind to it, allowing positively charged ions flow into the cells (Kumar, 2015). In order to test glyphosate competitive binding hypothesis, the glycine and the glutamic acid binding cavities from NMDA receptor (PDB ID: 4NF8) were analyzed.

The lower energy poses from docking analysis of glyphosate at

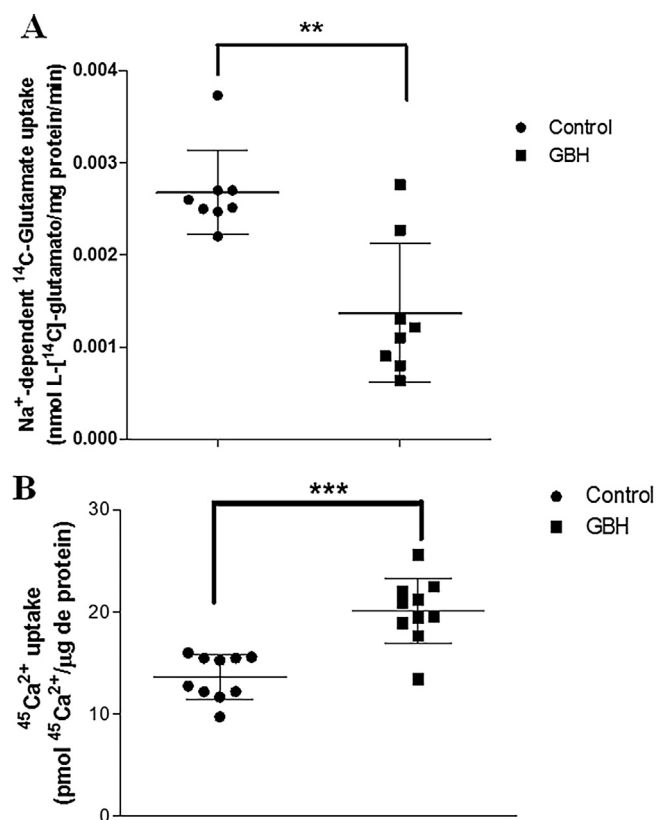


Fig. 2. Scatter plots of the effects of developmental exposure to GBH on glutamate neurotransmission and metabolism. The GBH exposure decreases [¹⁴C]-glutamate uptake (A) and increases ⁴⁵Ca²⁺ influx (B) in hippocampal slices from offspring on PND60. Data are reported as means ± S.D. Statistically significant differences from controls, as determined by Student's *t*-test were indicated: ***P* < 0.001 (comparison between variances *F* = 2.720, *p* = 0.2102); ****P* < 0.0001 (comparison between variances *F* = 2.196, *p* = 0.2568). The *F* test to compare variances showed that the variances are not significantly different, as specified.

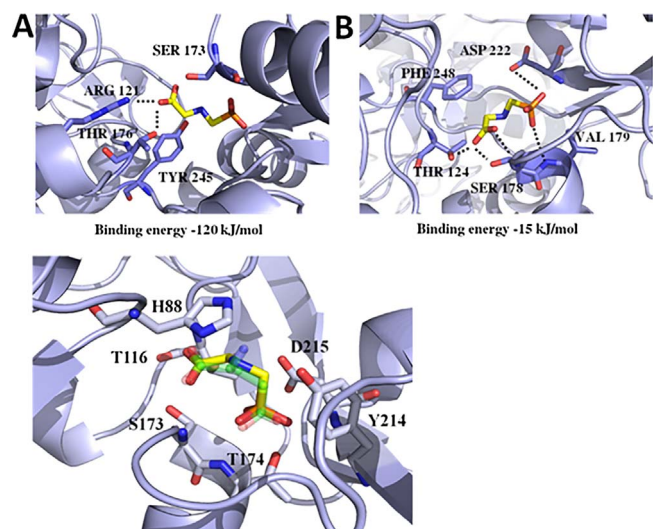


Fig. 3. Glyphosate binding to NMDA receptor (PDB ID: 4NF8). In (A), glyphosate into the glutamate binding cavity showing lower binding energy (−120.3 kJ/mol) compared to glycine binding cavity in (B) (−15.7 kJ/mol). Mapped residues involved in glyphosate binding are shown as sticks and glyphosate in yellow. Dashed lines represent polar contacts.

glutamic acid and glycine binding cavities of NMDA receptor were selected as starting points for molecular dynamic simulations and the binding energies were estimate during 12 ns molecular dynamic trajectories. The Fig. 3 shows the structure of glyphosate bound to the

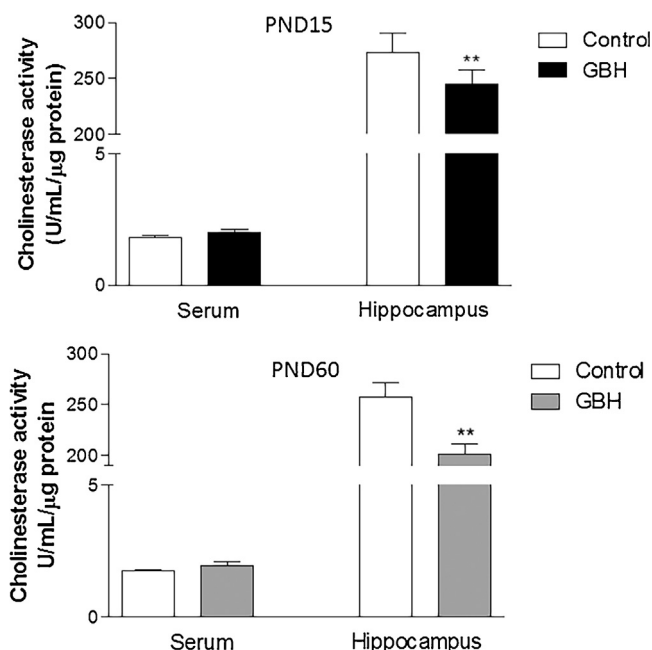


Fig. 4. Effects of developmental exposure to GBH on cholinesterase activity in hippocampal slices and serum from offspring on PND15 (A) and PND60 (B). Data are reported as means ± S.D. of 8 animals in each group. Statistically significant differences from controls, as determined by Student's *t*-test were indicated: ***P* < 0.01. The *F* test to compare variances showed that the variances are not significantly different.

glutamate binding cavities of NMDA receptor at 12 ns molecular dynamic simulations. The main amino acids involved in glyphosate binding at glutamate binding cavity (Chain B) were SER¹⁷³ and TYR²⁴⁵ (hydrophobic interactions), and ARG¹²¹ and THR¹¹⁶ (polar contacts with carboxyl group). At the glycine binding site (chain A), glyphosate interacts with PHE²⁴⁸ (hydrophobic interaction) and THR¹²⁴ and SER¹⁷⁸ (polar contacts with carboxyl group), VAL¹⁷⁹ and ASP²²² establishing polar contacts with phosphate group (Supplementary Fig. 1). The binding energy of glyphosate at glutamate and glycine binding cavities were −120 kJ/mol and −15 kJ/mol respectively. At glycine cavity the binding energy oscillates during dynamic simulation from positive to negative binding values suggesting less stability at the binding pocket compared to glutamate binding site (Supplementary Fig. 2).

3.4. Effects of GBH exposure on acetylcholinesterase activity

Our results also showed decreased acetylcholinesterase activity in hippocampus from immature (PND15) and adult (PND60) rats exposed to GBH. However, the activity of acetylcholinesterase was not altered by the herbicide exposure in blood from both immature and adult rats (Fig. 4A and 4B, respectively). These results demonstrate that even though we have not observed modifications in peripheral cholinesterase activity, glyphosate may alter cholinergic neurotransmission system by interfering in the acetylcholine metabolism.

3.5. S100B serum levels as a marker of astrocytic damage in GBH exposed immature and adult offspring

In an attempt to better understand the mechanisms involved in the neurotoxic damage induced by GBH exposure to immature and adult offspring brain, the levels of the astrocytic protein S100B were measured in blood of these animals. Results showed that GBH exposure decreased S100B serum levels on PND15 (Fig. 5A) and increase these levels on PND60 (Fig. 5B), suggesting that astrocyte dysfunction may be involved in the neurotoxic effects of the herbicide.

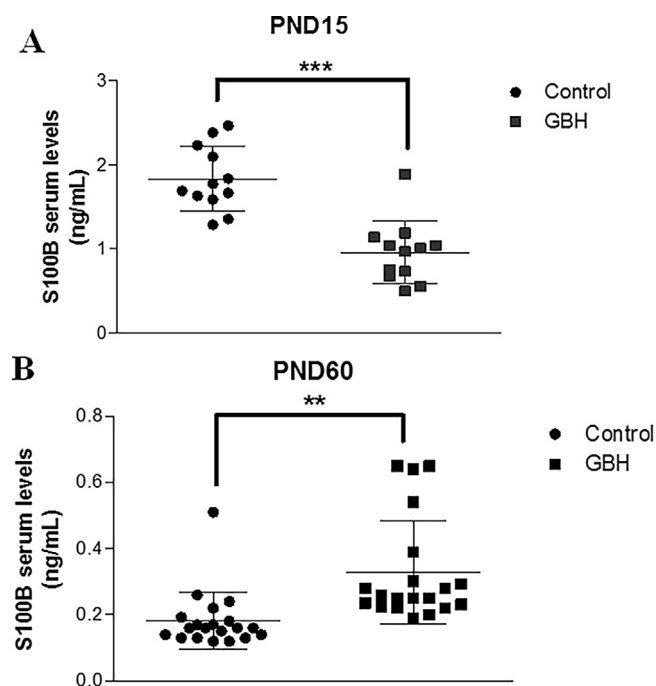


Fig. 5. Scatter plots showing the effects of developmental exposure to GBH on the levels of the astrocyte protein S100 B in serum of offspring on PND15 (A) and PND60 (B). S100 B immunocontent was determined by ELISA in serum of immature (N = 12) and adult rats (N = 20). Data are reported as means $\pm$ S.D. and expressed in ng/mL. Statistically significant differences from controls, as determined by unpaired Student's *t*-test or unpaired *t*-test with Welch's correction were indicated, respectively: ****P* < 0.0001 when the F test to compare variances showed that the variances are not significantly different (*F* = 1.061, *p* = 0.9237); ***P* < 0.001, the F test showed that variances are significantly different (*F* = 3.336, *P* = 0.0117).

3.6. Involvement of MAPK and NF- κ B pathways on the mechanism of GBH-induced neurotoxicity in rat hippocampus

Our previous studies demonstrated that the mechanism of GBH-induced toxicity involves the modulation of MAPK signaling pathways either in hippocampal (Cattani et al., 2014) or in testicular cells (de Liz Oliveira Cavalli et al., 2013). Moreover, it has been described that the mechanism of action of endocrine disruptors may involve the activation of cell signaling pathways, such as ERK (De Coster and van Larebeke, 2012). Supporting the involvement of MAPK pathway in GBH-induced neurotoxicity, Fig. 6A showed that maternal exposure to GBH increased the activated/phosphorylated levels of ERK1/2 in offspring rat hippocampus.

Considering that neurotoxic events are frequently associated with neuroinflammation aspects, we investigated the role of NF- κ B in the GBH-induced toxicity to immature rat hippocampus. Fig. 6B demonstrated that maternal GBH exposure leads to decreased phosphorylation of p65 of NF- κ B, which is in agreement with the activation of ERK1/2 that were probably activated as an attempt to revert the toxic events induced by the herbicide. Considering that NF- κ B transduction factors are abundant in brain and participate in diverse aspects of synaptic plasticity, learning and memory (Shih et al., 2015), the decreased levels of the activated protein may lead to modifications in the functions neuron and/or glial cells.

3.7. Effects of GBH exposure on biochemical parameters involved in oxidative damage in immature and adult rat hippocampus

In the present study, we have measured the consequences of maternal exposure to GBH to enzymatic and on enzymatic antioxidant defense system in offspring hippocampus on PND15 and PND60. Results showed that maternal exposure to the pesticide decreased the

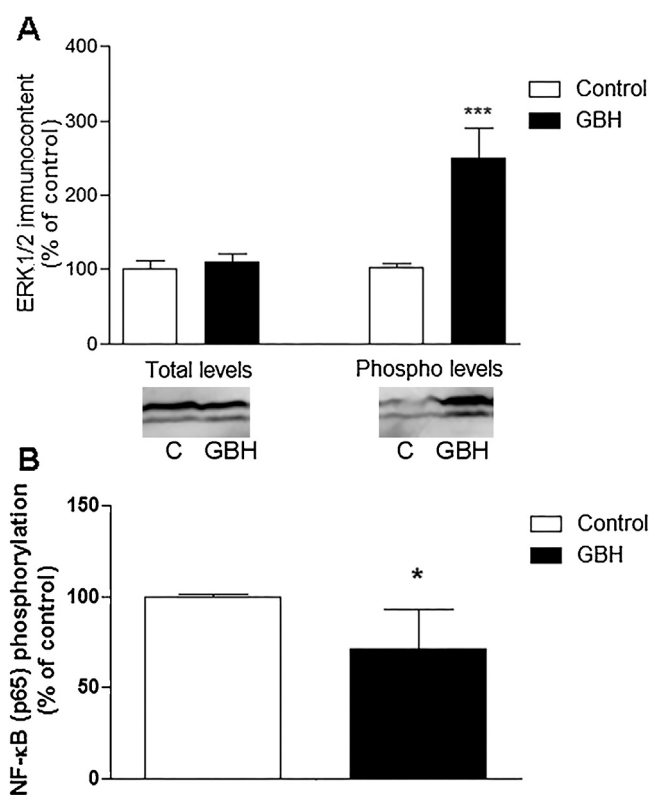


Fig. 6. Effect of maternal GBH exposure on ERK1/2 (A) and p65 NF- κ B (B) activation in rat hippocampus on PND15. Hippocampal slices were lysed and the total and phosphorylated levels of ERK1/2 and p65 NF- κ B were determined by Western blot (A) or ELISA (B). Data are reported as means $\pm$ S.D. for 8 animals in each group and expressed as % of Control. Statistically significant differences from controls, as determined by Student's *t* test are indicated: **P* < 0.05; ****P* < 0.0001 compared with control group. Representative immunoblots were shown in the insets. Ponceau staining of nitrocellulose membranes was used as loading control of immunoblots and total NF- κ B levels were used as controls in ELISA experiments. C = control; GBH = glyphosate-based herbicide. The F test to compare variances showed that the variances are not significantly different.

enzymatic activities of SOD, GST and G6PD SOD (Fig. 7A, E and F, respectively) as well as depleted GSH levels (Fig. 7H) in offspring hippocampus on PND15. On the other hand, the activity of GGT was induced by GBH exposure (Fig. 7G), while CAT, GR and GPx activities (Fig. 7B–D, respectively) were not altered by maternal exposure to herbicide in hippocampus from offspring pups on PND15. Fig. 8 showed the effects of maternal exposure to GBH on oxidative stress parameters in offspring hippocampus on PND60. Results showed that GBH exposure increased the activities of SOD, CAT and GST in hippocampus of adult offspring (Fig. 8A, B and 8E, respectively), while GR, GPx, G6PD and GGT enzymatic activities (Fig. 8C, D, F and G, respectively) were not altered by the pesticide exposure in this brain structure. In addition, the GSH levels were also not altered by GBH exposure in hippocampus from 60-day-old offspring (Fig. 8H). Taking together, our results clearly demonstrated that GBH exposure during development leads to oxidative damage in hippocampal cells.

3.8. Subchronic GBH exposure induces depressive-like behavior in adult rats

The present study also aimed to assess the association between GBH exposure during prenatal and postnatal life and neurobehavioral effects. The endpoints examined included neurochemical parameters and neurobehavioral outcomes. The effect of the subchronic exposure to GBH on the depressive-like behavior in offspring on PND60 was evaluated by the forced swimming test. Results showed that GBH exposure increased the immobility time in forced swimming test, which was associated with decreased climbing time, suggesting the induction of a

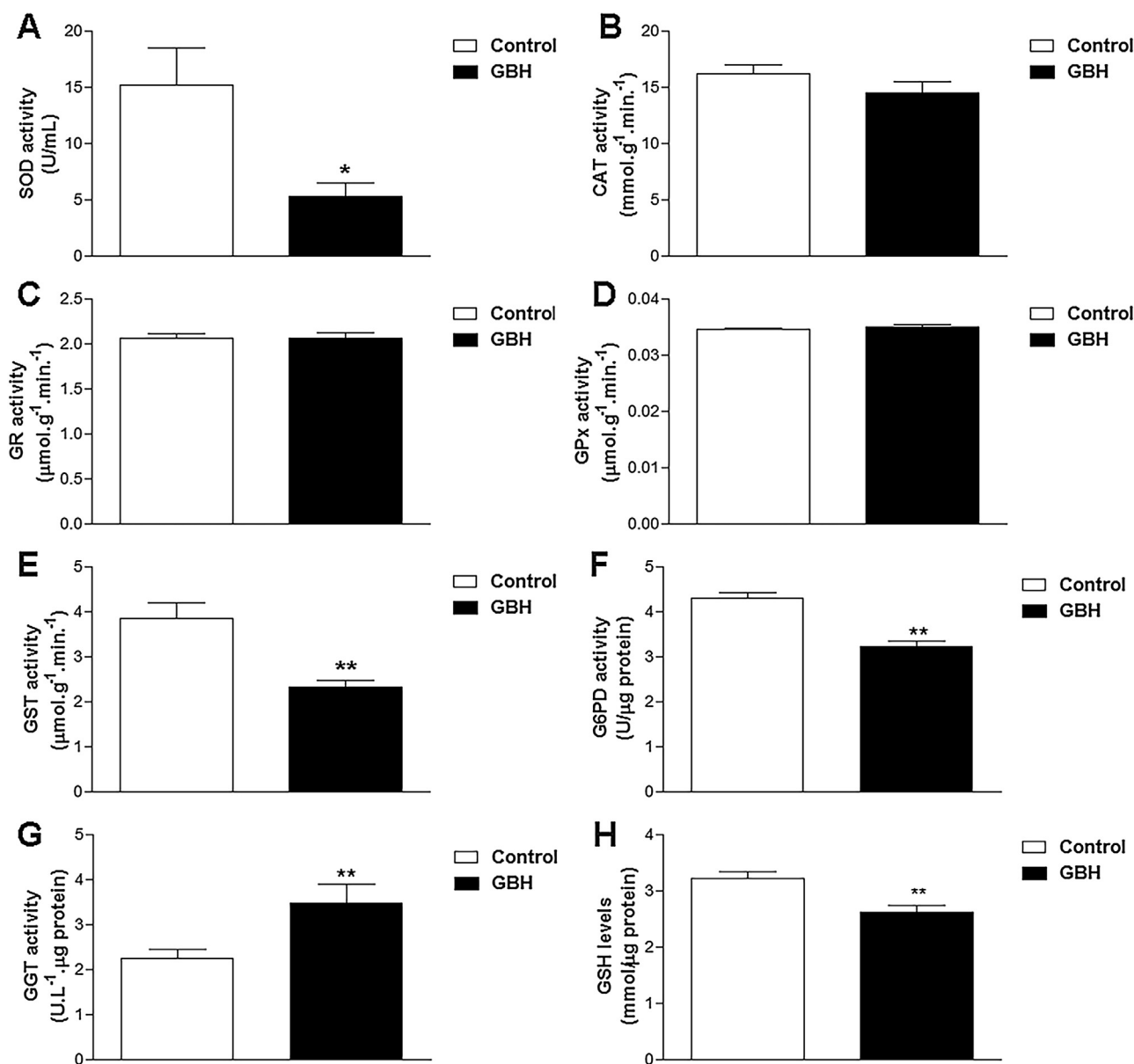


Fig. 7. Effects of maternal exposure to GBH on the enzymatic activity of superoxide dismutase (SOD), catalase (CAT), glutathione reductase (GR), glutathione peroxidase (GPx), glutathione-S-transferase (GST), glucose-6-phosphate dehydrogenase (G6PD) and gamma-glutamyl transferase (GGT), as well as on GSH levels in the hippocampus from immature offspring on PND15. Data are reported as means $\pm$ S.D. for 10 animals in each group. Statistically significant differences from controls, as determined by Student's *t*-test were indicated: **P* < 0.05; ***P* < 0.01. The *F* test to compare variances showed that the variances are not significantly different.

depressive-like profile (Fig. 9A and 9B).

The effects of the subchronic exposure with GBH on the anhedonia-like behavior evaluated in the sucrose consumption test are summarized in Fig. 9C. During the initial training phase, in which the rats were habituated to drink from two bottles (both filled with water), no significant difference was observed in water intake between the experimental groups. Finally, all groups consumed significantly more sucrose solution than water and no difference in total water intake during this time was observed.

The results of locomotor activity evaluated in the open field arena (Fig. 10A and 10B) and in the rotarod (Fig. 10C) revealed neither significant effects for the main factors in the total distance traveled (Fig. 10A), nor in the time spent in the central zone (Fig. 10B), or in the latency to fall (Fig. 10C).

4. Discussion

In the present study, we have shown that subchronic exposure to

GBH (from GD5 until PND60) leads to glutamate excitotoxicity, misregulation of cholinergic transmission, oxidative damage and astrocyte dysfunction in offspring hippocampus. The mechanisms underlying such effects involve the glyphosate interaction with NMDA receptor, overactivation of ERK and decreased activation of p65 NF- κ B signaling pathways. Moreover, these events were associated with depressive-like behavior observed in adult offspring (PND60).

It has been exhaustively demonstrated that the developing brain is more susceptible to the neurotoxic effects of pesticides than the adult brain. In addition, exposure to pesticides may cause significant neurobehavioral deficit, as demonstrated in humans and animal models (Bjorling-Poulsen et al., 2008; Modgil et al., 2014; London et al., 2012). Modgil et al. (2014) suggested that environmental exposure during prenatal and postnatal life may induce some epigenetic changes, which can lead to neurodegenerative conditions. In this context, it is necessary to consider that some consequences of early neurotoxic damages may not even emerge until advanced ages (Weiss, 2000).

Chronic exposure to pesticides has been associated with a wide

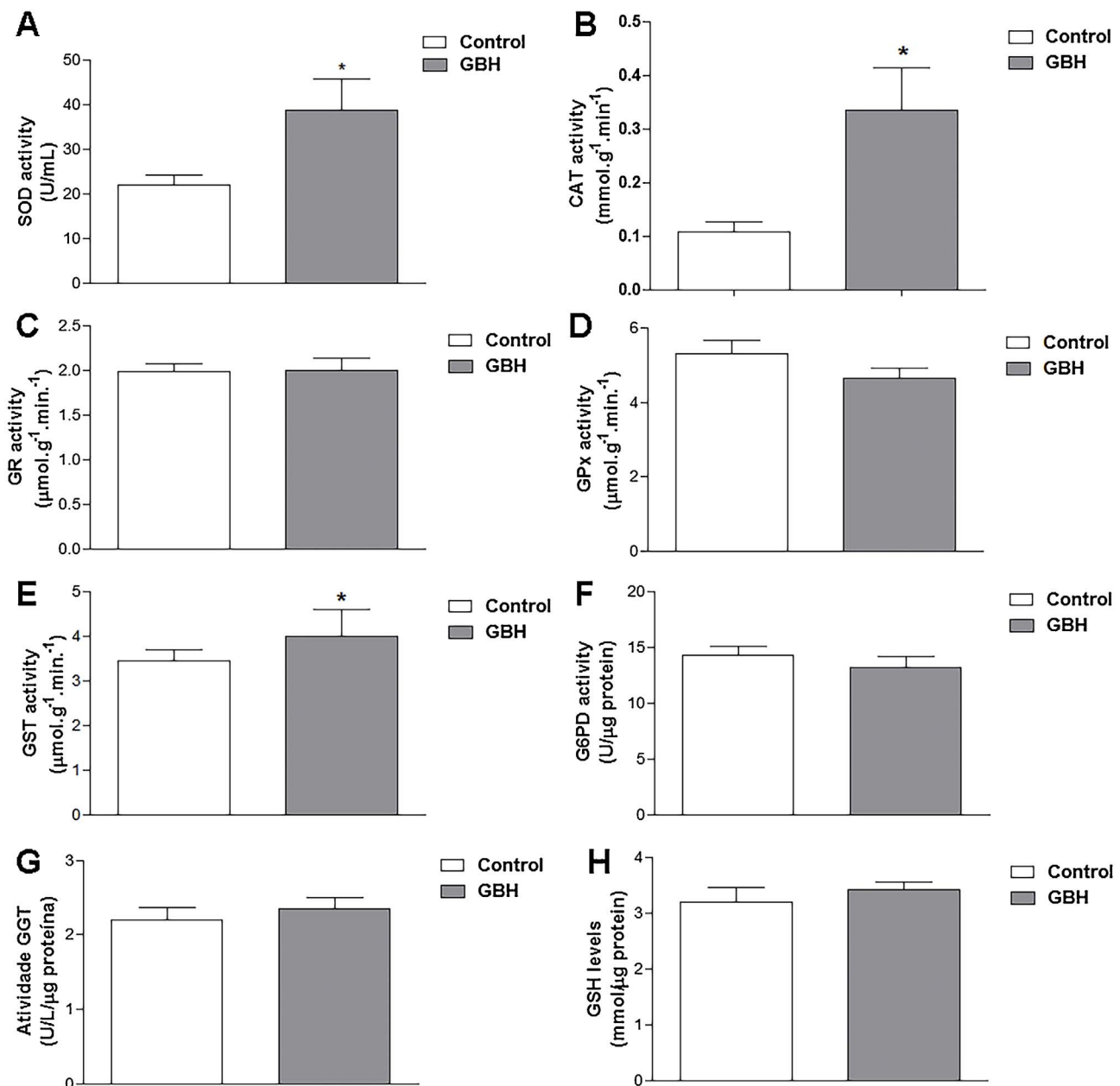


Fig. 8. Effect of developmental exposure to GBH on antioxidant parameters in hippocampus from adult offspring (PND60). The enzymatic activity of superoxide dismutase (SOD), catalase (CAT), glutathione reductase (GR), glutathione peroxidase (GPx), glutathione-S-transferase (GST), glucose-6-phosphate dehydrogenase (G6PD) and gamma-glutamyl transferase (GGT), as well as on GSH levels were measured in hippocampus from adult rats. Data are reported as means $\pm$ S.D. for 10 animals in each group. Statistically significant differences from controls, as determined by Student's *t*-test were indicated: **P* < 0.05. The *F* test to compare variances showed that the variances are not significantly different.

variety of neuronal disorders, and one of the best known is the possible association between pesticides exposure and Parkinson's Disease (Gatto et al., 2009; Li et al., 2016; Modgil et al., 2014; Parrón et al., 2011). Furthermore, there is evidence of increased prevalence of cognitive, neurobehavioral and psychomotor dysfunctions in individuals chronically exposed to pesticides (Zaganas et al., 2013).

Herein, we investigated the possible effects caused by subchronic *in vivo* exposure to GBH. The offspring exposition to GBH occurred from the gestational phase (GD5) throughout the adulthood (PND60). Male offspring at two different postnatal ages, PND15 and PND60, were used to evaluate early and late effects from glyphosate exposure through life.

The results of our study showed that GBH exposure did not change the average of delivery litter size as well as the number of males and females, but leads to decreased offspring body mass gain during different postnatal age. It is also important to consider that after delivery, the litter size was culled to eight pups in both control and treated

groups, indicating that the variability observed in offspring body mass gain was litter size-independent. Gallegos et al. (2016) also demonstrated that maternal exposure to GBH during gestation and suckling periods in rats did not affect delivery litter size. However, differently of our results, they showed no difference in offspring body mass gain at different postnatal ages.

In a previous study we have already showed that maternal exposure to GBH decreased glutamate uptake, increased glutamate release, overactivated NMDA receptors leading to an increased $^{45}\text{Ca}^{2+}$ influx in immature hippocampus from PND15 rats, demonstrating that GBH induced glutamatergic excitotoxicity phenomenon (Cattani et al., 2014). In this context, in the current study the same parameters were evaluated at PND60 to investigate whether these effects would persist until adulthood. The present results corroborate our previous findings showing that the decreased glutamate uptake and increased $^{45}\text{Ca}^{2+}$ influx earlier observed persisted until adulthood.

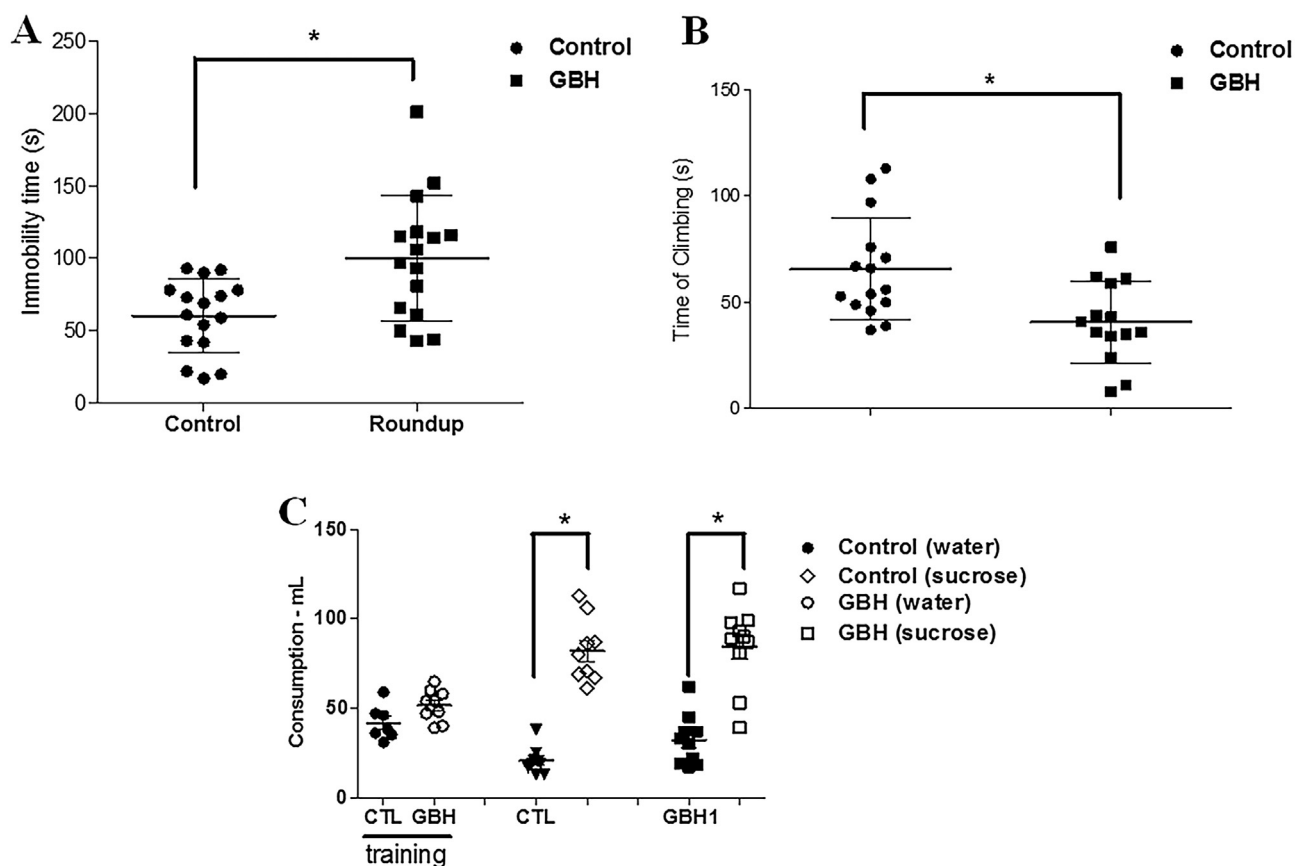


Fig. 9. Scatter plots of the effects of developmental exposure to GBH on depressive-like behavior and anhedonic-like behavior in adult offspring (PND60). The forced swimming test was used to measure the depressive-like behavior in adult offspring, where it was determined the time duration of immobility (A) and time of climbing (B). The sucrose preference test was used to measure the anhedonia-like behavior (C). Statis Data are reported as means $\pm$ S.D. tically significant differences from controls, as determined by unpaired Student's *t*-test or unpaired *t*-test with Welch's correction (A) were indicated, respectively: ***P* < 0.01.

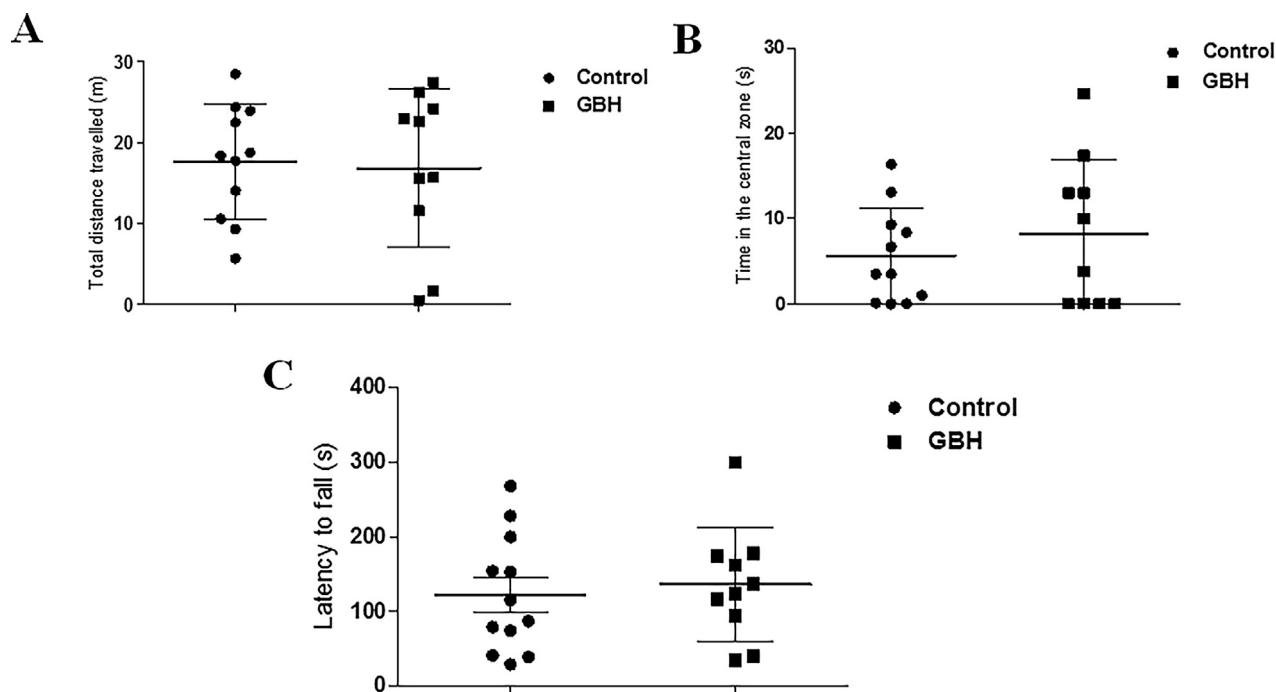


Fig. 10. Scatter plots of the effects of developmental exposure to GBH on locomotor activity of adult offspring (PND60). The spontaneous locomotor activity of adult rats was assessed in the open field test by measuring total distance travelled (A) and time in the central zone (B). The balance and motor coordination of rats were tested in the accelerated rotarod by measuring the latency to fall (C). Data are also expressed as means $\pm$ S.D. Results show that maternal GBH exposure do not lead to significant motor behavioral impairments in adult offspring.

Mechanism for the toxic effect of glyphosate on Ca^{2+} homeostasis was firstly demonstrated by Olorunsogo (Olorunsogo, 1990), which showed that this herbicide increased mitochondrial membrane permeability to protons and Ca^{2+} . In addition, we previously demonstrated that GBH-induced Ca^{2+} influx occurs through NMDA receptor activation and I-VDCC opening (Cattani et al., 2014).

Our previous findings over the glutamatergic system lead us to conclude that one of the mechanisms for the GBH-induced toxicity in rat hippocampus may involve the glutamatergic excitotoxicity phenomenon. Glutamate excitotoxicity is one of the most important mechanisms known of cell death in neurological diseases and consist of an increase in the intracellular calcium concentrations due to the over-activation of the NMDA glutamate receptor, which lead to an uncontrollable stimulation of proteolytic enzymes, lipoperoxidation and generation of ROS by mitochondria, culminating in programmed cell death (Arundine and Tymianski, 2003; Bacon et al., 1985; Fan and Raymond, 2007; Wang and Michaelis, 2010).

The NMDA receptors are heteromeric non-selective cation channels that require the binding of glycine and glutamate for gating. Considering the structural similarity of glyphosate with glutamate and glycine, we propose that the glutamate excitotoxicity induced by glyphosate might also be explained by its interaction/binding to NMDA glutamate receptors. Corroborating this hypothesis, molecular dynamic simulation results demonstrated that glyphosate may bind to glutamate and glycine cavities of NMDA receptors. The binding energies of glyphosate at the glutamate and glycine binding cavities, estimated during 12 ns molecular dynamic trajectories were -120 kJ/mol and -15 kJ/mol , respectively. Results also suggested lower stability of glyphosate binding at glycine binding pocket compared with glutamate one. Taking into account that NMDA receptor over-activation leads to excessive accumulation of intracellular calcium, the molecular dynamic results observed herein corroborate our previous study demonstrating that *in vitro* exposure to glyphosate (30 min) provoked calcium overload in hippocampal cells. The glyphosate-induced intracellular calcium accumulation was prevented by the use of NMDA antagonists (AP5 and MK801) as pharmacological tools. Thus, we concluded that glyphosate might interact with NMDA receptors and this event increase the Ca^{2+} ionic flow into cells, triggering mitochondrial dysfunction and oxidative damage.

In an attempt to investigate the effect of GBH on cholinergic neurotransmission, we determined the activity of acetylcholinesterase in offspring hippocampus. It is important to consider that peripheral cholinesterase activity is not a useful biomarker of glyphosate toxicity, which is in agreement with our results showing that the activity of this enzyme was not altered in serum from both immature and young adult GBH-exposed offspring. However, the cholinesterase activity in the hippocampus from immature and young adult-exposed offspring was decreased. A significant inhibitory effect on the activity of this enzyme by glyphosate was also observed by Menendez-Helman et al. (2012) in the fish *Cnesterodon decemmaculatus*, who proposed acetylcholinesterase as a biomarker of glyphosate-induced neurotoxicity in fish.

Acetylcholinesterase promotes acetylcholine degradation at neuromuscular junctions, this process is critical for both brain and muscle normal function. Enzymatic inhibition of cholinesterase induced by GBH leads to accumulation of acetylcholine in the synaptic cleft, which may cause a prolonged excitation of postsynaptic neurons causing an excitotoxic damage and cholinergic degeneration, as previously demonstrated (Zaganas et al., 2013). Taking this into account, ROS generation resulting from glutamatergic excitotoxicity and/or over-activation of the cholinergic system may be occurring in hippocampus from GBH-exposed offspring and might lead to subsequent deleterious oxidation of many important cellular biomolecules. These events may contribute to depressive-like behavior observed in offspring on PND60.

We previously showed that *in vitro* GBH exposure depletes GSH levels in immature rat hippocampus (Cattani et al., 2014). In the present

study we demonstrated that maternal exposure to GBH also decreased GSH levels on PND15, although we did not observed any changes in GSH concentration in hippocampus on PND60. This is expected considering that a systemic GSH depletion, which is very sensitive marker of oxidative stress, occurs after acute exposures to different contaminants, while it is usually recovered after few days (Halliwell and Gutteridge, 2007). We postulate that the different profile in GSH levels, observed in hippocampus between PND15 and PND60, occurs in order to maintain optimal levels of GSH within the young adult hippocampal cells via recycling the oxidized glutathione, or even through *de novo* glutathione synthesis (Halliwell and Gutteridge, 2007). An adaptive modification in antioxidant profile from PND15 to PND60 seems to occur, maintaining the levels of this important and ubiquitous non-enzymatic antioxidant GSH at basal concentrations in order to attenuate, at least in part, the toxic consequences of GBH to the brain.

The GST activity is an important defense strategy from cell against xenobiotics, beside its capability to also detoxify peroxides derived from lipoperoxidation processes (Halliwell and Gutteridge, 2007). Thus, decreased activity of this enzyme in hippocampus of immature animals may indicate insufficient detoxification of the pesticide associated with its initial inhibition. On the other hand, the increase in its activity in later-life (PND60) may be due to long-term treatment with GBH, where probably the expression of this enzyme was increased over long-term exposure in order to protect cells from the deleterious pesticide effects. Corroborating the decreased GST activity observed on PND15, a diminution in its activity was observed in tadpole after acute exposure to a commercial formulation of glyphosate (Lajmanovich et al., 2011).

Besides the decreased GSH levels, GBH-induced oxidative damage to immature rat hippocampal cells was also associated with a combined decrease in the activity of important antioxidant enzymes such as SOD, GST and G6PD. The activity of G6PD is involved in producing sufficient NADPH to maintain the cellular status of glutathione-cycle and keeping thiol groups of proteins and enzymes in the reduced state (Halliwell and Gutteridge, 2007). Thus, the decreased GSH levels observed on PND15, probably acting as first front antioxidant to directly counteract the contamination oxidative insult, might be also due to G6PD inhibition, which in turn, is essential for NADPH availability and for protection against oxidative stress. Corroborating the effect of GBH exposure on antioxidant defense system, results showed that CAT activity was unaltered in immature rat hippocampus despite a tendency to exhibit lower values, while it was increased in young adult brain. This finding can be explained by the fact that this enzyme had higher activity only after PND45 in rat brain (Shivakumar et al., 1991). Imbalance in the antioxidant enzyme activity and GSH concentration in hippocampus on PND15 may be indicative of an insufficient protection against ROS to face such oxidative insult or other physiological stressors in the developing brain. Studies have shown that the antioxidant capacity of the individual gradually increases during development from young to adult phase (Li et al., 2013), which could further explain the differences between antioxidant systems found in both ages.

In an attempt to understand the mechanism involved in the GBH-induced neurotoxicity, we demonstrated ERK1/2 overactivation associated with decreased NF- κ B phosphorylation in hippocampus of GBH-exposed rats. The ERK1/2 cascade has a wide range of substrates and can phosphorylate many target proteins in the cytosol or nucleus of neuronal cells, regulating important cellular functions such as proliferation, differentiation, apoptosis and synaptic plasticity (Nuttall and Oteiza, 2012). We previously demonstrated that GBH-induced toxicity in immature rat hippocampus after *in vitro* exposure also involves the recruitment of signal transduction pathways leading to the activation of kinase cascades including ERK (Cattani et al., 2014).

The NF- κ B signaling pathway activation is commonly associated with inflammatory process and unhealthy conditions. However, its activation is also involved in physiological events in brain such as neurogenesis (Koo et al., 2010), neuritogenesis (Rolls et al., 2007) and

synaptic plasticity (Ahn et al., 2008). Thus, the decreased NF- κ B phosphorylation and activation observed in immature rat hippocampus exposed to GBH during gestation and suckling periods might be affecting neurogenesis and consequently the development of central nervous system. Early life exposure to pesticide permethrin was able to induce a long-term brain changes included a decrease in the NF- κ B p65 protein level in hippocampus of 500 day-old Wistar rats (Carloni et al., 2013).

The modifications in the brain functions are also confirmed by changes in S100 B serum levels, which reflect the astroglial function. Our results showed that during development, the levels of this protein are decreased by GBH exposure. However, we observed increased S100 B levels in young adult offspring. We previously showed that maternal exposure to propylthiouracil, which leads to hypothyroidism, also increased S100 B serum levels (Cattani et al., 2013). Considering that the calcium-binding protein S100 B is considered an astrocyte marker, since these cells are the major S100B-containing compartment in central nervous system (Donato et al., 2013), our results suggest this protein as a peripheral biomarker of astrocyte dysfunction. A recent pilot study demonstrated that S100 B may serve as a marker for predicting neurologic complication in patients with glyphosate or glufosinate poisoning (Lee et al., 2017).

Due the increasing ratio between pesticide exposure and the onset of neurological disorders, we also investigated possible behavioral impairments resulting from subchronic exposure to GBH on PND60. Our results showed increased immobility time, which was associated with decreased climbing time in the forced swimming test, suggesting a depressive-like behavior in young adult rats subchronic exposed to GBH. Rats exposed to the pesticide malathion also feature the same characteristics in the forced swimming test (Brocardo et al., 2005). Chronic exposure to omethoate (organophosphorus pesticide) increased immobility time in behavioral tests and induced neuron damage in HE staining, inducing depressive-like behavior (Qiao et al., 2017).

Although the forced swimming test is best known for its use in validation drugs with potential antidepressant-like effects (Porsolt et al., 1978), it can also be used to evaluate depressive potential of some compounds (Paul et al., 1990). In this model, rats are forced to swim in a cylinder from which they cannot escape and the increase in the immobility time is used as a depressive-like behavior indicator, since it is considered that the animal gave up trying to escape. Based on these validated protocols, we postulate that subchronic exposure to GBH provoked a depressive-like behavior in young adult offspring. However, pesticide exposure did not lead to anhedonia-like behavior in young adult offspring, as demonstrated by sucrose consumption test.

The open field and rota-rod tests were used to evaluate the consequences of GBH exposure on locomotor activity. Both tests showed no signs of locomotor deficits in GBH-exposed animals compared to controls. These results enable the finding obtained in the forced swimming test, since if the locomotor activity is decreased it could be generating a false positive result for depressive-like behavior. On the other hand, Hernandez-Plata et al. (2015) described effects of glyphosate exposure on locomotor activity. The authors reported that repeated exposure to glyphosate caused hypoactivity immediately after each injection, and it was also apparent 2 days after the last injection in rats exposed to the highest dose but not after 16 days, suggesting that these alterations occur while a “critical” concentration of glyphosate is present in the system.

Our results demonstrated that subchronic exposure to GBH might result in outcomes ranging from oxidative damage to impairment of brain function, which may account for depressive-like behavior later in life. Considering that GBH are the widely used herbicide, we suggested that the depressive-like behavior induced by early exposure to this herbicide, as observed in the present study, might be involved, at least in part, in the elevated risk of suicide observed among farmers (Bossard et al., 2016; Porrón et al., 2011; Freire and Koifman, 2013; Tiesman et al., 2015).

The hippocampus is one of several limbic structures that have been extensively studied in individuals with depression. Glutamatergic excitotoxicity phenomenon is thought to be one of the causes of depressive disorder and as such is considered to be a potential pharmacological target for the treatment of depression. Administration of antagonists of the glutamatergic system, including zinc and ketamine, has shown an antidepressant effect in preclinical as well as clinical studies (Mlyniec, 2015; Sahay and Hen, 2007; Miladinovic et al., 2015; Huang et al., 2017).

In our experimental model, we supposed that the offspring are exposed to the pesticide that cross the placental barrier during gestational period (Mose et al., 2008; Poulsen et al., 2009) and/or that is given by the mother through breast milk during suckling period. The exposure of the dams (GD5 to PND21) and young adult offspring (PND22 to PND60) to the pesticide is through the ingestion of drinking water containing GBH.

Although its oral bioavailability is considered low, glyphosate is slowly eliminated from plasma and therefore would be diffused to target tissues to exert systemic effects (Anadon et al., 2009). Since glyphosate has an amino acid-like structure, the affinity of this herbicide for the amino acid transporters suggests that additional toxicities may be exhibited in organs expressing high amino acid transporter activity (Xu et al., 2016), such as brain. Indeed, the detection of glyphosate in brain (Menkes et al., 1991) and cerebrospinal fluid (Sato et al., 2011) after human exposure to GBH indicating that this herbicide is able to cross the blood brain barrier.

The present study demonstrated that subchronic GBH exposure leads to the glutamatergic excitotoxicity phenomenon with increased $^{45}\text{Ca}^{2+}$ influx and oxidative damage in hippocampus from both immature (PND15) (Cattani et al., 2014) and young adult animals (PND60). The increase in intracellular calcium concentration can be related to an increased ROS production, promoting an imbalance in the cellular antioxidant defense system, which is dependent on the duration of exposure. Furthermore, our results show that oxidative damage may be occurring during the neurodevelopmental period.

5. Conclusions

The GBH-induced neurotoxicity after subchronic exposure (from GD5 until PND60) involves glutamate excitotoxicity phenomenon due to increased glutamate release into synaptic cleft, decreased glutamate uptake and glyphosate interaction with NMDA receptors, thereby leading to ionic flow of Ca^{2+} into hippocampal cells. Such event culminates in oxidative stress, astrocyte dysfunction and depressive-like behavior. Some of the mechanisms underlying the GBH-induced neurotoxicity involve the NMDA receptor activation, S100 B release, ERK1/2 phosphorylation and decreased p65 NF- κ B activation. Moreover, the acetylcholinesterase inhibition also suggests the involvement of the cholinergic system, in addition to the glutamatergic one, in the neurotoxicity induced by GBH in rat hippocampus. Further studies covering other neurotransmitter systems are necessary to clarify all the mechanisms involved in GBH-induced neurotoxicity.

Declaration of interest

Authors declare that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research here reported.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tox.2017.06.001>.

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